

			ADNOC GAS HABSHAN & BAB CONTROL SYSTEMS UPGRADE (PACKAGE 1) PROJECT NO. 0405323						  			
			COMMENTS RESOLUTION SHEET									
Document Title:			PIPING AND INSTRUMENTATION DIAGRAM GAS SEAL SYSTEM DIAGRAM FOR 13-C-101A/B				AGP Response Code			Prepared By:	VK	
Document No:			5323P1-H1-013-00-50-002		Revision No:	B	1. Revise and Re-submit		3. Approved	Checked By:	DV	
AG Ref. No:			ADNOCGP-WTRAN-007250		Date:	6-Jun-2024	2. Approved with comments	✓	4. For Information only	Approved By:	YVR	
SN	Reference		COMPANY Comments				CONSULTANT Response			COMPANY Resolution		
	Page No.	Clause No.										
1	1	Process Comment	Please review there is a change between Rev.A and Rev.B Changes shall be mentioned in the CRS or notes (typical).				Noted, However all the pressure alarm values are updated as per operating manual.					
2	-	-					Following changes are made in P&ID w.r.t. Rev.B submission as per piping/instrumentation inputs: 1. Root valves for all PT/DPT/FT are shown in project scope cloud based on instrument input. 2. Control valve PDCV-3640 Tie-in location updated based on piping input. No piping tie-in required on inlet line to PDCV-3640 as these are tubing. Hence, Tie-in number TP-810 A/B & 4016 A/B are updated as 13-TP-810 & 13-TP-4016. 3. Note 28 and 29 added. 4. Table to mention instruments in project scope for 13-C-101A and 13-C-101B is added. Refer detail "B".					

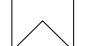
NOTICE: THIS DRAWING AND THE INFORMATION IT CONTAINS IS THE SOLE PROPERTY OF ADNOC GAS PROCESSING IT SHALL NOT BE REPRODUCED OR LENT WITHOUT THE AGREEMENT OF THE OWNER

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12-Aug-2024 idm260310

TEBODIN M.E. LTD.

NOTES:-

- FOR COMPONENT DESCRIPTION SEE BILL OF MATERIAL 478-120-201 AND 478-120-202.
- FOR CUSTOMER CONNECTION LETTER DESCRIPTION SEE DRAWING 478-145-601.
- FOR PIPING LINE CODE EXPLANATION SEE DRAWING 003-152-001.
- FOR COMPONENT SYMBOLS AND LETTER DESIGNATIONS SEE DRAWING 478-373-601.
- RUN VENT LINES OUTSIDE OF BUILDING AS SHOWN. DO NOT CONNECT VENTS TOGETHER EXCEPT AS SHOWN ON THIS DRAWING.
- PRESSURE IN THIS VENT LINE MUST BE LIMITED AT ALL TIMES SO THAT MAXIMUM SEAL BACK PRESSURE DOES NOT EXCEED 0.35 BARG (5 PSIG.)
-  - LABRINTH
 - GAS SEAL
 - CARBON RING SEAL
- A FLOW ORIFICE REQUIRES 20 PIPE DIAMETERS OF STRAIGHT PIPE UPSTREAM AND 10 PIPE DIAMETERS OF STRAIGHT PIPE DOWNSTREAM.
- CLEAN DRY SEPARATION NITROGEN SUPPLY 7 BAR G(100 PSIG), 2.8 BAR G(40 PSIG) MIN. FILTERED TO 3 MICRON ABSOLUTE AND 99.97% LIQUID FREE.
- SEAL GAS SUPPLY (SWEET FUEL GAS) CLEAN DRY FILTERED TO 10 MICRON PARTICLE (SOLIDS) AND 99.97% LIQUID FREE OF PARTICLES 3 MICRONS & LARGER.
- FOR WIRING DETAILS SEE DRAWING 478-223-201.
- REFERENCE LUBE OIL DIAGRAM 317-201-201.
- SAFETY LABEL-TOXIC VAPORS (HYDROGEN SULFIDE GAS) IS TO BE AFFIXED TO GAS SEAL PANEL SEE DWG. 016-289-001.

HOLDS:-

- DELETED.
- DELETED.

LEGENDS:-

	SCOPE CLOUD		XX - UNIT NUMBER TP - TIE-IN POINT
	PANEL MOUNTED		YYY - SERIAL NUMBER
	MOUNTED BEHIND THE PANEL		VALVE - INST. OR GLOBE - NOR. OPEN
	TURBINE CONTROL SYSTEM		VALVE - INST. OR GLOBE - NOR. CLOSED
	REVISION CLOUD		
	LOCAL MOUNTED		
	DCS		

REFERENCE:-

- NM-5323P1-H1-013-15-21-001 - INSTRUMENT INDEX - UNIT 13
- 03-00-30001 TO 007 - LEGEND FOR HABSHAN-1
- NM-5323P1-H1-01-00-40-002 - ALARM AND TRIP SETTINGS LIST - HABSHAN 1

USE THIS DRAWING FOR INFORMATION GIVEN WITHIN CLOUDED AREA

THIS DRAWING IS DEVELOPED FROM ADNOC DRAWING NO. 478-120 SHT 1/2 & SHT 2/2 REV 04 (PACKAGE / VENDOR DRAWING)

CONTINUATION OF NOTES :

- CASE CONNECTION DESCRIPTION
A2-1-SEAL GAS SUPPLY 1-1/2"-900# ANSI RTJ FLANGE (INTAKE & DISCH.)
B1-1-INNER (PRIMARY) SEAL VENT 1-1/2"-900# ANSI RTJ FLANGE (INTAKE & DISCH.)
C1-1-OUTER (SECONDARY) SEAL VENT 1-1/2"-900# ANSI RTJ FLANGE (INTAKE & DISCH.)
D1-1-SEAL REFERENCE PRESSURE 1-1/2"-900# ANSI RTJ FLANGE (INTAKE & DISCH.)
F1-1-SEPARATION GAS INJECTION 1-1/2"-900# ANSI RTJ FLANGE (INTAKE & DISCH.)
H2-1-LUBE OIL VENT 2"-150# ANSI RF FLANGE (INTAKE & DISCH.)
H1-1-LUBE OIL DRAIN 4"-150# ANSI RF FLANGE (INTAKE & DISCH.)
N1-1-BALANCE CONNECTION 3"-900# ANSI RTJ FLANGE (INTAKE & DISCH.)
N2-1-BALANCE GAGE CONNECTION 1"-900# ANSI RTJ FLANGE - D-E
G1-1-LUBE OIL SUPPLY (NOT SHOWN) 2"-150# ANSI RF FLANGE (INTAKE & DISCH.)
- CASE DRAINS TO BE PIPED SEPARATELY TO EDGE OF SKID.
- BALANCE LINE CONNECTIONS TO BE BALANCE LINE DIFFERENTIAL PRESSURE GAUGE (PDI-3638) MOUNTED ON COMPRESSOR SKID.
- SWEET FUEL GAS SEAL GAS SUPPLY SHUTOFF VALVE (SUPPLIED BY CUSTOMER) IS REQUIRED TO SHUT OFF SEAL GAS WHEN UNIT IS TRIPPED SINCE GAS SUPPLY IS NOT FROM COMPRESSOR DISCHARGE.
- ALL INSTRUMENT TAG NUMBERS SHOWN ARE FOR THE "FEED GAS" TRAIN "A" AND ARE PREFIXED WITH 13- (EX. 13-PDT-3631). TAG NUMBERS FOR TRAIN-B ARE PREFIXED WITH 13- (EX. 13-PDT-3731).
- ALL SHUTDOWNS ARE "AND" LOGIC UNLESS OTHERWISE NOTED.
- FOR GAS SEAL DESCRIPTION OF OPERATION SEE 478-120-601.
- EXISTING INSTRUMENT/VALVES ARE REPLACED WITH NEW INSTRUMENT/VALVES AS PER PROJECT (0405323) SCOPE.
- FOR 13-C-101A, CORRESPONDING EXISTING INSTRUMENT ARE REPLACED WITH NEW INSTRUMENT AS PER PROJECT (0405323) SCOPE.
- NEW INSTRUMENT ARE INSTALLED AS PER PROJECT (0405323) SCOPE.
- ALL PIPING /INSTRUMENT INTERFACE CONNECTION WILL BE AS PER HOOK-UP DIAGRAM (5323P1-H1-01-15-38-001 & 5323P1-H1-01-15-38-002).
- FOR TRAIN-B (13-C-101B), EXISTING INSTRUMENT 13-PT-3734B IS RETAINED.
- THE TURBINE CONTROL SYSTEM (TCS) SHALL BE INTERFACED TO DCS THROUGH REDUNDANT MODBUS TCP/IP COMMUNICATION.
- THE EXISTING UNIT CONTROL PANEL (UCP) RELATED TO GE RX 3I AND WOODWARD IS REPLACED WITH NEW TURBINE CONTROL SYSTEM.
- INSTRUMENT TUBING TO BE RECONNECTED TO THE PIPING AFTER PIPE MODIFICATION.
- REFER DETAIL "B" FOR LIST OF INSTRUMENTS/VALVES, WHICH ARE REPLACED OR ADDED AS PER PROJECT (0405323) SCOPE.

DETAIL-"A"		
PACKAGE	INSTRUMENT TAG	TIE-IN NO.
13-C-101A	13-PDCV-3639	13-TP-809 A/B
13-C-101B	13-PDCV-3739	13-TP-4015 A/B
13-C-101A	13-PDCV-3640	13-TP-810
13-C-101B	13-PDCV-3740	13-TP-4016

NOTE-29 DETAIL-"B"		
13-C-101A	13-C-101B	
13-FT-1625	13-FT-1628	
13-FT-1725	13-FT-1728	
13-PDCV-3632	13-PDCV-3732	
13-PDCV-3639	13-PDCV-3739	
13-PDCV-3640	13-PDCV-3740	
13-PDT-3631	13-PDT-3731	
13-PDT-3632A	13-PDT-3732A	
13-PDT-3633	13-PDT-3733	
13-PDY-3639	13-PDY-3739	
13-PT-3634A	13-PT-3734A	
13-PT-3634B		
13-PT-3635A	13-PT-3735A	
13-PT-3635B	13-PT-3735B	
13-PT-3636A	13-PT-3736A	
13-PT-3636B	13-PT-3736B	
13-PZT-3639	13-PZT-3739	

CONTRACT NO. 21712C/D	BY	DATE	GAS SEAL
TYPE INST	DRAWN	KJS	07/06/92
SCALE NONE	CHECKED	GMW	07/07/92
DEPT. 63.81	REDRAW		
DRAFTING APPROVAL	D.L.A.	07/07/92	TURBO PRODUCTS DIVISION
CONTRACT APPROVAL	JAB	07/07/92	CLEAN, N.Y. USA
SYSTEM APPROVAL	GMW	07/07/92	
LICENSEE			

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CONTRACT APPROVAL	JAB	07/07/92	CLEAN, N.Y. USA
SYSTEM APPROVAL	GMW	07/07/92	
LICENSEE			

0	09.08.2024	ISSUED FOR CONSTRUCTION (IFC)	PRA	VK/DV	YVR	WAM
B	27.05.2024	ISSUED FOR APPROVAL (IFA)	PRA	VK/DV	YVR	WAM
A	13.01.2024	ISSUED FOR COMPANY REVIEW (ICR)	PT	KG/TS	YVR	WAM
REV	DATE	DESCRIPTION	DRAWN	CHKD	APPRD	APPRD (TARGET)
SCALE: NTS	DRAWN: PRA	CHKD: VK/DV	APPRD: YVR	APPRD (TARGET): WAM		

CONTRACTOR:	ADNOC GAS	ENGINEERING SUBCONTRACTOR:	BILFINGER
PROJECT TITLE:	HABSHAN & BAB CONTROL SYSTEMS UPGRADE PACKAGE 1		
DRAWING TITLE:	PIPING AND INSTRUMENTATION DIAGRAM GAS SEAL SYSTEM DIAGRAM FOR 13-C-101A/B		

UNIT: 13	PLANT: HABSHAN-1
CONTR. PROJ. No.: AET01074.00	CONT. DWG. No.: 3143222
ADNOC GAS PROJ. No.: 0405323	ADNOC GAS DWG. No.: 5323P1-H1-013-00-50-002
SHEET No.: 01/01	REV: 0
SHEET SIZE: A1	